A 6-DOF Hand-Eye Calibration Method Based on Position Residuals for Micromanipulation

Haoran Cui, Weichen Dou, Jia Shi, Bowen Wang, Yuhan Chen, *Member, IEEE*, and Xinyu Liu, *Member, IEEE*

***Abstract*—Accurate hand-eye calibration with six-degrees-of-freedom (6-DOF) in micromanipulation remains a significant challenge. The limited field of view and shallow depth of field hinder multi-feature tracking and orientation estimation, while substantially lower axial localization precision results in anisotropic measurement noise. To address these limitations, this article proposes a position-residual-based invariant extended Kalman filter (PR-IEKF). The method estimates the complete hand-eye transformation from successive 3-D position measurements of a single tracked feature and known robot motions, without requiring orientation measurements. Specifically, an analytical Jacobian is derived to relate the position residual to the hand-eye transformation error. Based on this Jacobian, motion excitation conditions are analyzed to establish sufficient constraints for full 6-DOF estimation. Finally, an invariant extended Kalman filter incorporates the residual model and anisotropic measurement covariance to recursively update the hand-eye transformation through multiplicative corrections on the $SE(3)$ manifold. Simulations demonstrate the effectiveness of the proposed method, and experiments with three objective magnifications show a 9.9% to 17.6% reduction in mean lateral motion-consistency error relative to position-only least squares. MATLAB and C++ implementations are publicly available.**

***Index Terms*—Hand-eye calibration, micromanipulation, machine vision, Lie-group filtering, anisotropic noise.**

I. Introduction

In micromanipulation systems, hand-eye calibration establishes the spatial transformation between the microscopic vision frame and the robot base frame, and calibration accuracy directly affects the positioning accuracy of vision-guided operations [1], [2]. Many micromanipulation systems use visual feedback primarily within the focal plane [1]. With the development of applications such as photonic chip packaging [3] and micro-electro-mechanical systems assembly [4], more micromanipulation tasks require position and orientation adjustment involving out-of-plane motion. Therefore, reliable six-degree-of-freedom (6-DOF) hand-eye calibration is increasingly important for micromanipulation.

The difficulty of extending hand-eye calibration beyond the focal plane is compounded by limited pose observability and anisotropic measurement noise [1], [5]. The shallow depth of field limits the reliable recovery of out-of-plane pose information. In practice, axial position can be estimated using autofocus-based methods [6]. However, axial localization generally exhibits substantially higher uncertainty than lateral localization [1], [5], resulting in anisotropic measurement noise.

Many classical hand-eye calibration methods rely on 6-DOF pose information. These methods commonly formulate the calibration problem as rigid-motion equations such as $\boldsymbol{AX} = \boldsymbol{XB}$ and $\boldsymbol{AX} = \boldsymbol{YB}$, where $\boldsymbol{A}$ and $\boldsymbol{B}$ are known rigid-body transformations derived from robot kinematics and visual pose measurements. A broad range of solution techniques has been developed for these formulations, including separable estimation [7], Lie-group methods [8], dual-quaternion formulations [9], and optimization-based calibration [10]. Related hand-eye calibration formulations have also been extended to more complex vision-guided robotic systems, including viewpoint selection in robotic vision [11], eye-in-hand 3-D scanner-to-robot calibration [12], and multi-eye–multi-hand robotic systems [13]. Other methods reduce the dependence on full visual pose measurements by incorporating point correspondences [14] or known geometric constraints [15], [16]. Despite their different formulations, these methods generally require visual orientation information, multiple point correspondences, or additional geometric constraints.

Position-only hand-eye calibration provides an alternative when reliable orientation measurements are unavailable. Such methods have been developed for conventional robotic systems [17], [18] and estimate the hand-eye transformation from 3-D point observations and known robot motions using position-only least squares (PO-LS). Although they avoid direct orientation measurements, they do not explicitly account for the anisotropic measurement noise encountered in microscopic imaging. In addition, the linear rotation estimate is not constrained to lie on $SO(3)$ and therefore requires a subsequent projection.

In micromanipulation, limited out-of-plane observability is commonly addressed through two strategies. One strategy

This work was supported in part by the National Key R&D Program of China under Grant 2023YFB4706700, in part by the National Natural Science Foundation of China under Grants 62503076, 62588301, and 62433006, in part by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China under Grant JYB2025XDXM117, and in part by the Doctoral Scientific Research Foundation of Liaoning Province in China under Grant 2026-BS-0054. (Corresponding authors: Yuhan Chen and Xinyu Liu.)

Haoran Cui, Weichen Dou, Jia Shi, Bowen Wang, and Yuhan Chen are with the Institute of Robotics and Intelligent Systems, Dalian University of Technology, Dalian 116024, China (e-mail: cuihaoran@mail.dlut.edu.cn; 20211043040@mail.dlut.edu.cn; sj2024@mail.dlut.edu.cn; wangbowen999@mail.dlut.edu.cn; yhchen@dlut.edu.cn).

Xinyu Liu is with the Institute of Robotics and Intelligent Systems, Dalian University of Technology, Dalian 116024, China, and also with the Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, ON M5S 3G8, Canada (e-mail: xyliu@mie.utoronto.ca).

restricts calibration and visual feedback to the focal plane to avoid direct axial measurement. Examples include monocular visual control [19] and planar robot-to-image calibration in retinal microsurgery [20]. These approaches simplify visual measurement but generally estimate only an in-plane transformation, leaving the out-of-plane components of the hand-eye transformation unresolved. The other strategy introduces additional sources of spatial information. For example, stereo or multi-view microscopic systems recover 3-D information from multiple optical views [21], and microscope-integrated optical coherence tomography (OCT) provides volumetric information for needle-tip localization and hand-eye calibration [22]. Spatial information can also be estimated directly from microscopic images using data-driven models, such as methods for pose and depth estimation [23], or introduced through geometric priors. For example, computer-aided design models of the end effector have been used for position and pose estimation under microscopic imaging [4]. Although these approaches provide additional spatial information, they typically require additional sensing hardware, data-driven models, or accurate geometric priors.

To address these limitations, this article proposes a position-residual-based invariant extended Kalman filter (PR-IEKF) for 6-DOF hand-eye calibration in micromanipulation. PR-IEKF estimates the hand-eye transformation from successive 3-D position measurements of a single tracked feature and known commanded robot motions, without direct orientation measurements. An analytical Jacobian relates the position residual to the hand-eye error, and an anisotropic position measurement covariance accounts for the unequal uncertainty in lateral and axial measurements. The hand-eye transformation is recursively updated on $SE(3)$. The main contributions of this article are summarized as follows:

1) A position-residual-based method is developed for 6-DOF hand-eye calibration from successive 3-D position observations of a single tracked feature and known commanded robot motions, without direct orientation measurements.
2) An analytical Jacobian is derived without numerical differentiation to relate the position residual to the hand-eye error, enabling estimation of both rotation and translation under sufficient motion excitation.
3) PR-IEKF incorporates an anisotropic position measurement covariance and recursively updates the hand-eye transformation on $SE(3)$.
4) The proposed method is validated through simulations and experiments on a micromanipulation platform with $5\times$, $10\times$, and $20\times$ objectives. The MATLAB and C++ implementations are publicly available at https://github.com/Qininining/OMHEC.

The remainder of this article is organized as follows. Section II formulates the calibration problem, and Section III presents the proposed PR-IEKF. Sections IV and V present the simulation and experimental evaluations, respectively. Finally, Section VI concludes the article and discusses future work.

II. Problem Formulation

This section formulates the 6-DOF hand-eye calibration problem using known robot motions and successive 3-D position observations of a single tracked feature.

A. Mathematical Preliminaries

Let $\boldsymbol{T} \in SE(3)$ denote a rigid-body transformation,

$$\boldsymbol{T} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{t} \\ \boldsymbol{0}^\top & 1 \end{bmatrix}, \tag{1}$$

where $\boldsymbol{R} \in SO(3)$ and $\boldsymbol{t} \in \mathbb{R}^3$ denote the rotation matrix and translation vector, respectively.

For a spatial twist $\boldsymbol{\xi} = [\boldsymbol{\omega}^\top, \boldsymbol{v}^\top]^\top \in \mathbb{R}^6$ over a sampling interval Δt, the corresponding exponential coordinates are

$$\boldsymbol{\lambda} = \boldsymbol{\xi}\Delta t = \begin{bmatrix} \boldsymbol{\phi} \\ \boldsymbol{\rho} \end{bmatrix}, \tag{2}$$

where $\boldsymbol{\phi}$ and $\boldsymbol{\rho}$ denote the rotational and translational components, respectively. The wedge operator maps $\boldsymbol{\lambda}$ to $\mathfrak{se}(3)$ as

$$\boldsymbol{\lambda}^\wedge = \begin{bmatrix} [\boldsymbol{\phi}]_\times & \boldsymbol{\rho} \\ \boldsymbol{0}^\top & 0 \end{bmatrix}, \tag{3}$$

where $[\cdot]_\times$ denotes the skew-symmetric operator. The exponential and logarithm maps satisfy

$$\boldsymbol{T} = \exp(\boldsymbol{\lambda}^\wedge), \qquad \boldsymbol{\lambda} = (\log(\boldsymbol{T}))^\vee, \tag{4}$$

where $(\cdot)^\vee$ denotes the inverse of the wedge operator.

For $\boldsymbol{T} \in SE(3)$, the Lie-group adjoint and Lie-algebra adjoint are defined as

$$\mathrm{Ad}_{\boldsymbol{T}} = \begin{bmatrix} \boldsymbol{R} & \boldsymbol{0} \\ [\boldsymbol{t}]_\times \boldsymbol{R} & \boldsymbol{R} \end{bmatrix}, \qquad \mathrm{ad}_{\boldsymbol{\lambda}} = \begin{bmatrix} [\boldsymbol{\phi}]_\times & \boldsymbol{0} \\ [\boldsymbol{\rho}]_\times & [\boldsymbol{\phi}]_\times \end{bmatrix}. \tag{5}$$

They satisfy

$$\mathrm{Ad}_{\exp(\boldsymbol{\lambda}^\wedge)} = \exp(\mathrm{ad}_{\boldsymbol{\lambda}}). \tag{6}$$

The above definitions provide the Lie-group and Lie-algebra results used in Section II-C; further details can be found in [24], [25].

B. Microscopic Measurement Model

1) Geometric Model: Fig. 1 illustrates the coordinate frames and the position-residual construction used for hand-eye calibration. The robot base frame $\{b\}$ is fixed to the stationary base of the robot, and the end-effector frame $\{e\}$ is attached to the moving platform. The microscope frame $\{c\}$ is defined at the reference focal plane corresponding to $q = 0$, where q denotes the axial displacement used during autofocus. The origin of $\{c\}$ is located at the image center. The x_c and y_c axes lie in the focal plane, and the z_c axis is aligned with the optical axis. The feature point $\boldsymbol{p}$ is rigidly attached to the target and is tracked throughout the robot motion.

The hand-eye transformation maps coordinates from frame $\{b\}$ to frame $\{c\}$. Its true value and estimate at step k are denoted by ${}^c\boldsymbol{T}_b^*$ and ${}^c\hat{\boldsymbol{T}}_{b,k}$, respectively. The 3-D position of the tracked feature in the microscope frame is denoted by

$${}^c\boldsymbol{p}_k = \begin{bmatrix} x_{c,k} & y_{c,k} & z_{c,k} \end{bmatrix}^\top. \tag{7}$$

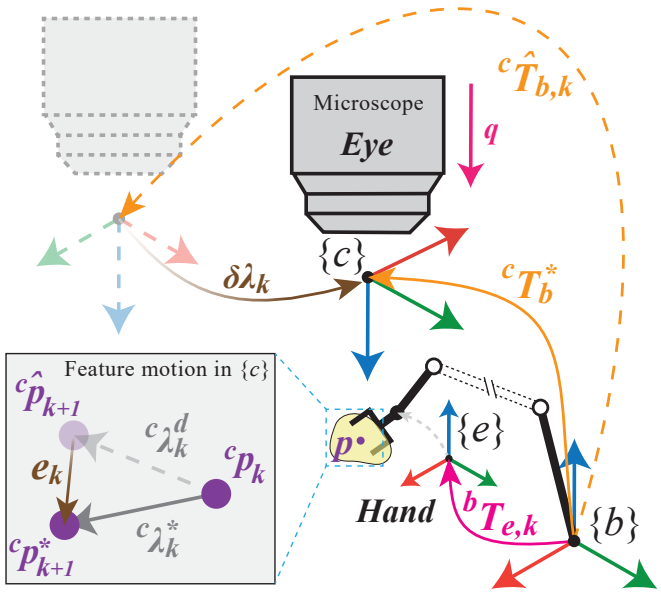


Fig. 1. Coordinate frames and position-residual construction for hand-eye calibration.

The lateral coordinates (x_c, y_c) are obtained from image-plane localization using a calibrated image scale. The axial coordinate z_c is estimated by autofocus. Specifically, an image sequence $I(q)$ is acquired over a set of sampled axial positions $\mathcal{Q}$, and the best-focus position is determined as

$$q^* = \arg\max_{q\in\mathcal{Q}} \mathcal{F}\big(I(q)\big), \tag{8}$$

where $\mathcal{F}(\cdot)$ is a gradient-energy focus measure constructed by summing the squared horizontal and vertical image gradients [26]. With the reference focal plane defined at $q = 0$ and the axis convention in Fig. 1, the axial coordinate is given by $z_c = q^*$.

2) Anisotropic Measurement Model: Axial localization by autofocus generally exhibits higher measurement noise than lateral image localization [5], [27], [26], resulting in anisotropic position measurement noise. The measured feature position is modeled as

$$^c\tilde{\boldsymbol{p}} = {}^c\boldsymbol{p} + \boldsymbol{n}, \qquad \boldsymbol{n} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_p), \tag{9}$$

where $\boldsymbol{n}$ denotes the zero-mean measurement noise. Assuming independent measurement errors along the three axes of frame $\{c\}$, the position measurement covariance is

$$\boldsymbol{\Sigma}_p = \mathrm{diag}\left(\sigma_x^2, \sigma_y^2, \sigma_z^2\right), \tag{10}$$

with

$$\sigma_z^2 \gg \sigma_x^2 \approx \sigma_y^2. \tag{11}$$

This covariance model characterizes the anisotropic measurement noise and is incorporated into the recursive estimator in Section III-C.

C. Position Residual Definition

As illustrated in Fig. 1, consider one calibration step from k to $k+1$. For the feature position ${}^c\boldsymbol{p}_k$ before the motion, consider a desired motion ${}^c\boldsymbol{\lambda}_k^d$ in the microscope frame, defined as

$$^c\boldsymbol{\lambda}_k^d = \begin{bmatrix} {}^c\boldsymbol{\phi}_k^d \\ {}^c\boldsymbol{\rho}_k^d \end{bmatrix} \in \mathbb{R}^6. \tag{12}$$

Under this desired motion, the predicted feature position at step $k+1$ is denoted by ${}^c\hat{\boldsymbol{p}}_{k+1}$.

The current hand-eye estimate ${}^c\hat{\boldsymbol{T}}_{b,k}$ generally differs from the true transformation ${}^c\boldsymbol{T}_b^*$. Their error is represented by

$$\delta\boldsymbol{\lambda}_k = \Big(\log\Big({}^c\boldsymbol{T}_b^*({}^c\hat{\boldsymbol{T}}_{b,k})^{-1}\Big)\Big)^\vee. \tag{13}$$

Consequently, the actual motion in frame $\{c\}$, denoted by ${}^c\boldsymbol{\lambda}_k^*$, differs from the desired motion ${}^c\boldsymbol{\lambda}_k^d$, resulting in the actual feature position ${}^c\boldsymbol{p}_{k+1}^*$. The corresponding position residual is defined as

$$\boldsymbol{e}_k = {}^c\boldsymbol{p}_{k+1}^* - {}^c\hat{\boldsymbol{p}}_{k+1}. \tag{14}$$

This residual characterizes the difference between the actual and predicted feature positions in the microscope frame.

D. Problem Statement

For each calibration step from k to $k+1$, the available quantities are the robot pose ${}^b\boldsymbol{T}_{e,k}$ obtained from robot kinematics, the desired motion ${}^c\boldsymbol{\lambda}_k^d$ defined in (12), and the measured feature positions ${}^c\tilde{\boldsymbol{p}}_k$ and ${}^c\tilde{\boldsymbol{p}}_{k+1}$ satisfying (9). The corresponding measurement uncertainty is characterized by $\boldsymbol{\Sigma}_p$ in (10).

The true hand-eye transformation ${}^c\boldsymbol{T}_b^*$ remains constant during the calibration sequence and is unknown. At step k, its current estimate is ${}^c\hat{\boldsymbol{T}}_{b,k}$, with the transformation error $\delta\boldsymbol{\lambda}_k$ defined in (13). For the desired motion ${}^c\boldsymbol{\lambda}_k^d$, the current estimate ${}^c\hat{\boldsymbol{T}}_{b,k}$ determines the predicted feature position ${}^c\hat{\boldsymbol{p}}_{k+1}$, and the true transformation ${}^c\boldsymbol{T}_b^*$ determines the actual motion ${}^c\boldsymbol{\lambda}_k^*$ and the resulting feature position ${}^c\boldsymbol{p}_{k+1}^*$. Their difference defines the position residual $\boldsymbol{e}_k$ in (14).

Therefore, given the desired motions ${}^c\boldsymbol{\lambda}_k^d$, measured feature positions ${}^c\tilde{\boldsymbol{p}}_k$ and ${}^c\tilde{\boldsymbol{p}}_{k+1}$, measurement covariance $\boldsymbol{\Sigma}_p$, and an initial hand-eye estimate ${}^c\hat{\boldsymbol{T}}_{b,0}$, the calibration problem is to estimate the constant hand-eye transformation ${}^c\boldsymbol{T}_b^* \in SE(3)$.

III. Proposed Calibration Framework

Based on Section II, this section presents the PR-IEKF framework, including analytical Jacobian derivation, motion excitation analysis, and recursive estimation on $SE(3)$ with anisotropic measurement covariance.

A. Analytical Jacobian Derivation

To estimate the unknown hand-eye transformation ${}^c\boldsymbol{T}_b^*$ defined in Section II-D, the position residual $\boldsymbol{e}_k$ is related to the hand-eye transformation error $\delta\boldsymbol{\lambda}_k$ and can be locally expressed as

$$\boldsymbol{e}_k = \boldsymbol{J}_k\delta\boldsymbol{\lambda}_k + \mathbf{o}_k, \tag{15}$$

where $\boldsymbol{J}_k \in \mathbb{R}^{3\times 6}$ is the Jacobian relating the position residual to the hand-eye transformation error, and $\mathbf{o}_k$ denotes the higher-order residual terms. By estimating $\delta\boldsymbol{\lambda}_k$, the current hand-eye transformation ${}^c\hat{\boldsymbol{T}}_{b,k}$ can be updated toward the true transformation ${}^c\boldsymbol{T}_b^*$.

1) Feature Position Prediction: For the desired motion ${}^c\boldsymbol{\lambda}_k^d$ defined in (12) and the feature position ${}^c\boldsymbol{p}_k$, the predicted feature position is

$$\begin{bmatrix} {}^c\hat{\boldsymbol{p}}_{k+1} \\ 1 \end{bmatrix} = \exp\left(({}^c\boldsymbol{\lambda}_k^d)^\wedge\right) \begin{bmatrix} {}^c\boldsymbol{p}_k \\ 1 \end{bmatrix}. \tag{16}$$

Expanding the exponential term in

$$\exp\left(({}^c\boldsymbol{\lambda}_k^d)^\wedge\right) = \boldsymbol{I}_4 + ({}^c\boldsymbol{\lambda}_k^d)^\wedge + \mathcal{O}\left(\|{}^c\boldsymbol{\lambda}_k^d\|_2^2\right), \tag{17}$$

where $\mathcal{O}(\cdot)$ collects the higher-order terms, and substituting (17) into (16) gives

$${}^c\hat{\boldsymbol{p}}_{k+1} = {}^c\boldsymbol{p}_k + \boldsymbol{M}({}^c\boldsymbol{p}_k){}^c\boldsymbol{\lambda}_k^d + \mathcal{O}\left(\|{}^c\boldsymbol{\lambda}_k^d\|_2^2\right), \tag{18}$$

where

$$\boldsymbol{M}({}^c\boldsymbol{p}_k) = \begin{bmatrix} -[{}^c\boldsymbol{p}_k]_\times & \boldsymbol{I}_3 \end{bmatrix}. \tag{19}$$

2) Position-Residual Relation: Using ${}^c\hat{\boldsymbol{T}}_{b,k}$, the desired motion ${}^c\boldsymbol{\lambda}_k^d$ can be expressed in frame $\{b\}$ as

$${}^b\boldsymbol{\lambda}_k^d = \mathrm{Ad}_{({}^c\hat{\boldsymbol{T}}_{b,k})^{-1}}{}^c\boldsymbol{\lambda}_k^d. \tag{20}$$

The resulting ${}^b\boldsymbol{\lambda}_k^d$ is used to determine the commanded robot pose. Given the current robot pose ${}^b\boldsymbol{T}_{e,k}$, the commanded pose at step $k+1$ is

$${}^b\boldsymbol{T}_{e,k+1}^d = \exp\left(({}^b\boldsymbol{\lambda}_k^d)^\wedge\right){}^b\boldsymbol{T}_{e,k}. \tag{21}$$

The actual motion ${}^c\boldsymbol{\lambda}_k^*$ in the microscope frame $\{c\}$ is obtained from the commanded robot motion ${}^b\boldsymbol{\lambda}_k^d$ through the true hand-eye transformation ${}^c\boldsymbol{T}_b^*$ as

$${}^c\boldsymbol{\lambda}_k^* = \mathrm{Ad}_{{}^c\boldsymbol{T}_b^*}{}^b\boldsymbol{\lambda}_k^d = \mathrm{Ad}_{{}^c\boldsymbol{T}_b^*({}^c\hat{\boldsymbol{T}}_{b,k})^{-1}}{}^c\boldsymbol{\lambda}_k^d. \tag{22}$$

Using (13), this relation can be written as

$${}^c\boldsymbol{\lambda}_k^* = \mathrm{Ad}_{\exp(\delta\boldsymbol{\lambda}_k^\wedge)}{}^c\boldsymbol{\lambda}_k^d. \tag{23}$$

Using the adjoint identity in (6), the adjoint transformation can be expanded as

$$\mathrm{Ad}_{\exp(\delta\boldsymbol{\lambda}_k^\wedge)} = \boldsymbol{I}_6 + \mathrm{ad}_{\delta\boldsymbol{\lambda}_k} + \mathcal{O}\left(\|\delta\boldsymbol{\lambda}_k\|_2^2\right). \tag{24}$$

Substituting (24) into (23) gives

$${}^c\boldsymbol{\lambda}_k^* = {}^c\boldsymbol{\lambda}_k^d + \mathrm{ad}_{\delta\boldsymbol{\lambda}_k}{}^c\boldsymbol{\lambda}_k^d + \mathcal{O}\left(\|\delta\boldsymbol{\lambda}_k\|_2^2\|{}^c\boldsymbol{\lambda}_k^d\|_2\right). \tag{25}$$

Following (18), the feature position after the actual increment ${}^c\boldsymbol{\lambda}_k^*$ is

$${}^c\boldsymbol{p}_{k+1}^* = {}^c\boldsymbol{p}_k + \boldsymbol{M}({}^c\boldsymbol{p}_k){}^c\boldsymbol{\lambda}_k^* + \mathcal{O}\left(\|{}^c\boldsymbol{\lambda}_k^*\|_2^2\right). \tag{26}$$

Substituting (18) and (26) into (14) gives

$$\boldsymbol{e}_k = \boldsymbol{M}({}^c\boldsymbol{p}_k)\left({}^c\boldsymbol{\lambda}_k^* - {}^c\boldsymbol{\lambda}_k^d\right) + \mathcal{O}\left(\|{}^c\boldsymbol{\lambda}_k^*\|_2^2 + \|{}^c\boldsymbol{\lambda}_k^d\|_2^2\right). \tag{27}$$

Combining (27) with (25) yields

$$\boldsymbol{e}_k = \underbrace{-\boldsymbol{M}({}^c\boldsymbol{p}_k)\,\mathrm{ad}_{{}^c\boldsymbol{\lambda}_k^d}}_{\boldsymbol{J}_k}\delta\boldsymbol{\lambda}_k + \mathbf{o}_k, \tag{28}$$

where

$$\boldsymbol{J}_k = \underbrace{\left[[{}^c\boldsymbol{p}_k]_\times[{}^c\boldsymbol{\phi}_k^d]_\times - [{}^c\boldsymbol{\rho}_k^d]_\times\right.}_{\boldsymbol{J}_{\phi,k}} \underbrace{\left.-[{}^c\boldsymbol{\phi}_k^d]_\times\right]}_{\boldsymbol{J}_{\rho,k}}. \tag{29}$$

The Jacobian is therefore obtained analytically from the feature position and the desired motion increment, without numerical differentiation.

B. Motion Excitation Analysis

According to (29), the Jacobian $\boldsymbol{J}_k \in \mathbb{R}^{3\times 6}$ consists of $\boldsymbol{J}_{\phi,k}$ and $\boldsymbol{J}_{\rho,k}$, which characterize the sensitivities of the position residual to the rotational and translational hand-eye errors, respectively. Since a single motion provides only three position constraints, multiple motions are required to constrain all six components of the hand-eye transformation error.

For N motions, the Jacobians are stacked as

$$\boldsymbol{\mathcal{J}}_{1:N} = \begin{bmatrix} \boldsymbol{J}_1^\top & \cdots & \boldsymbol{J}_N^\top \end{bmatrix}^\top \in \mathbb{R}^{3N\times 6}. \tag{30}$$

The condition for full local constraints on the six hand-eye error components is

$$\mathrm{rank}\left(\boldsymbol{\mathcal{J}}_{1:N}\right) = 6. \tag{31}$$

The required motion excitation can be analyzed from the structure of $\boldsymbol{J}_{\phi,k}$ and $\boldsymbol{J}_{\rho,k}$. For the translational error, $\boldsymbol{J}_{\rho,k} = -[{}^c\boldsymbol{\phi}_k^d]_\times$. Since the skew-symmetric matrix $[\boldsymbol{a}]_\times$ has rank two for any nonzero $\boldsymbol{a} \in \mathbb{R}^3$, a single rotational motion cannot constrain the translational-error component along its rotation axis. Rotations about non-collinear axes are therefore required to constrain all translational-error components.

For the rotational error, under pure translation, $\boldsymbol{J}_{\phi,k} = -[{}^c\boldsymbol{\rho}_k^d]_\times$. Thus, a single translational motion cannot constrain the rotational-error component along its translation direction. Translations along non-collinear directions provide complementary constraints on the rotational error components.

Therefore, sufficiently diverse rotational and translational motions are required to satisfy (31) and constrain all six components of the hand-eye transformation error.

C. IEKF-Based Recursive Estimation

The analytical Jacobian derived in Section III-A relates the position residual to the hand-eye transformation error. Based on this relation, an invariant extended Kalman filter (IEKF) is used to recursively estimate the hand-eye transformation on $SE(3)$.

Let $\boldsymbol{X} = {}^c\boldsymbol{T}_b^* \in SE(3)$ denote the unknown hand-eye transformation. At step k, $\hat{\boldsymbol{X}}_k^-$ and $\hat{\boldsymbol{X}}_k^+$ denote the prior and posterior estimates, respectively. Consistent with (13), the right-invariant error is defined as

$$\boldsymbol{\eta}_k^R = \boldsymbol{X}\left(\hat{\boldsymbol{X}}_k^-\right)^{-1} = \exp\left(\delta\boldsymbol{\lambda}_k^\wedge\right). \tag{32}$$

where $\delta\boldsymbol{\lambda}_k \in \mathbb{R}^6$ is the corresponding Lie-algebra error. For the estimated correction $\delta\hat{\boldsymbol{\lambda}}_k \in \mathbb{R}^6$, the manifold update is

$$\hat{\boldsymbol{X}}_k^+ = \exp\left(\delta\hat{\boldsymbol{\lambda}}_k^\wedge\right)\hat{\boldsymbol{X}}_k^-. \tag{33}$$

This keeps the estimate on $SE(3)$ while expressing the correction in the local Lie algebra. Using the measured feature positions, define the residual as

$$\tilde{\boldsymbol{e}}_k = {}^c\tilde{\boldsymbol{p}}_{k+1} - {}^c\hat{\boldsymbol{p}}_{k+1}, \tag{34}$$

and

$$\tilde{\boldsymbol{e}}_k = \boldsymbol{H}_k \delta\boldsymbol{\lambda}_k + \boldsymbol{v}_k, \tag{35}$$

where $\boldsymbol{v}_k \in \mathbb{R}^3$ is the residual noise. The measurement Jacobian $\boldsymbol{H}_k$ is obtained from (29) by evaluating the analytical Jacobian at the measured feature position:

$$\boldsymbol{H}_k = \boldsymbol{J}_k|_{{}^c\boldsymbol{p}_k = {}^c\tilde{\boldsymbol{p}}_k} = -\boldsymbol{M}({}^c\tilde{\boldsymbol{p}}_k)\,\mathrm{ad}_{{}^c\boldsymbol{\lambda}_k^d}. \tag{36}$$

Let $\boldsymbol{R}_k^d = \exp([{}^c\boldsymbol{\phi}_k^d]_\times) \in SO(3)$ denote the rotational component of ${}^c\boldsymbol{\lambda}_k^d$. Since the residual combines two independent position measurements, its covariance is

$$\boldsymbol{\Sigma}_{e,k} = \boldsymbol{\Sigma}_p + \boldsymbol{R}_k^d \boldsymbol{\Sigma}_p \left(\boldsymbol{R}_k^d\right)^\top, \tag{37}$$

which propagates the anisotropic position measurement uncertainty into the PR-IEKF update. Consecutive residuals share one position observation. To retain the standard sequential IEKF structure, each update uses the corresponding marginal residual covariance in (37) without explicitly maintaining inter-residual covariance.

Let $\boldsymbol{P}_k^- \in \mathbb{R}^{6\times6}$ denote the prior covariance of the hand-eye error and $\boldsymbol{Q} \in \mathbb{R}^{6\times6}$ the process covariance. The Kalman gain and correction are

$$\boldsymbol{K}_k = \boldsymbol{P}_k^- \boldsymbol{H}_k^\top \left(\boldsymbol{H}_k \boldsymbol{P}_k^- \boldsymbol{H}_k^\top + \boldsymbol{\Sigma}_{e,k}\right)^{-1}, \tag{38}$$

$$\delta\hat{\boldsymbol{\lambda}}_k = \boldsymbol{K}_k \tilde{\boldsymbol{e}}_k. \tag{39}$$

The posterior covariance is updated in Joseph form [28]:

$$\begin{aligned}\boldsymbol{P}_k^+ = {} & (\boldsymbol{I}_6 - \boldsymbol{K}_k\boldsymbol{H}_k)\,\boldsymbol{P}_k^-\,(\boldsymbol{I}_6 - \boldsymbol{K}_k\boldsymbol{H}_k)^\top \\ & + \boldsymbol{K}_k \boldsymbol{\Sigma}_{e,k} \boldsymbol{K}_k^\top.\end{aligned} \tag{40}$$

Because the hand-eye transformation is constant during calibration, $\boldsymbol{Q} = \boldsymbol{0}$, and the prediction step reduces to

$$\hat{\boldsymbol{X}}_{k+1}^- = \hat{\boldsymbol{X}}_k^+, \qquad \boldsymbol{P}_{k+1}^- = \boldsymbol{P}_k^+. \tag{41}$$

Thus, the position residual, analytical Jacobian, and anisotropic residual covariance are integrated into the recursive IEKF update.

D. Summary of the PR-IEKF Framework

The preceding derivations establish the position-residual measurement model, the excitation condition for full 6-DOF estimation, and the analytical Jacobian used in the recursive update. Together with the anisotropic residual covariance, these components form the complete PR-IEKF framework on $SE(3)$. The method requires only the commanded robot motion and successive 3-D feature positions, and recursively estimates the hand-eye transformation from the resulting position residuals. The complete implementation procedure is summarized in Algorithm 1. The source code for the proposed method is publicly available at https://github.com/Qinininning/OMHEC.

Algorithm 1 PR-IEKF for 6-DOF Hand-Eye Calibration

Input: Initial estimate $\hat{\boldsymbol{X}}_0^-$, initial covariance $\boldsymbol{P}_0^-$, position measurement covariance $\boldsymbol{\Sigma}_p$, initial measurement ${}^c\tilde{\boldsymbol{p}}_0$, and excitation sequence $\{{}^c\boldsymbol{\lambda}_k^d\}_{k=0}^{N-1}$

Output: Estimated hand-eye transformation $\hat{\boldsymbol{X}}_{N-1}^+$

for $k = 0, 1, \ldots, N-1$ **do**
  Obtain ${}^c\boldsymbol{\lambda}_k^d$ from the excitation sequence
  Read the current robot pose ${}^b\boldsymbol{T}_{e,k}$
  Transform ${}^c\boldsymbol{\lambda}_k^d$ to ${}^b\boldsymbol{\lambda}_k^d$ using (20)
  Predict ${}^c\hat{\boldsymbol{p}}_{k+1}$ using (18) with ${}^c\boldsymbol{p}_k = {}^c\tilde{\boldsymbol{p}}_k$
  Compute the commanded pose ${}^b\boldsymbol{T}_{e,k+1}^d$ using (21)
  Command the robot to move to ${}^b\boldsymbol{T}_{e,k+1}^d$
  Acquire ${}^c\tilde{\boldsymbol{p}}_{k+1}$
  Compute $\tilde{\boldsymbol{e}}_k$ and $\boldsymbol{H}_k$ using (34) and (36)
  Set $\hat{\boldsymbol{X}}_k^+ \leftarrow \hat{\boldsymbol{X}}_k^-$ and $\boldsymbol{P}_k^+ \leftarrow \boldsymbol{P}_k^-$
  Compute $\boldsymbol{\Sigma}_{e,k}$ and $\boldsymbol{K}_k$ using (37) and (38)
  Compute $\delta\hat{\boldsymbol{\lambda}}_k$ using (39)
  Update $\hat{\boldsymbol{X}}_k^+$ and $\boldsymbol{P}_k^+$ using (33) and (40)
  if $k < N-1$ **then**
    Predict $\hat{\boldsymbol{X}}_{k+1}^-$ and $\boldsymbol{P}_{k+1}^-$ using (41)
  end if
end for

IV. Simulation

This section evaluates the proposed PR-IEKF under varying initialization, excitation, and measurement noise conditions to demonstrate its advantages over existing methods.

A. Simulation Setup

A fixed hand-eye transformation ${}^c\boldsymbol{T}_b^* \in SE(3)$ was used as the ground truth. Each simulation condition was evaluated over 100 Monte Carlo runs, with each run consisting of $N = 100$ calibration motion increments. The rotational and translational motion step sizes were $\|{}^c\boldsymbol{\phi}_k^d\|_2 = 1^\circ$ and $\|{}^c\boldsymbol{\rho}_k^d\|_2 = 0.10$ mm, respectively. The motion directions were selected such that the stacked Jacobian formed by the first six motion increments had rank six, satisfying the full-rank condition in (31). The initial hand-eye error was set to $\|\delta\boldsymbol{\phi}_0\|_2 = 10^\circ$ and $\|\delta\boldsymbol{\rho}_0\|_2 = 1.0$ mm. The measurement noise standard deviations were $(\sigma_x, \sigma_y, \sigma_z) = (0.003, 0.003, 0.030)$ mm. For PR-IEKF, the initial covariance was constructed using rotation and translation scales of 30° and 0.8 mm, respectively.

The proposed PR-IEKF was compared with PO-LS [17], [18] and position-residual fixed-gain descent (PR-FGD)[29]. The former is a position-only least-squares method in which the rotation estimate is subsequently projected onto $SO(3)$. The latter uses the same measured position residual and measurement Jacobian as PR-IEKF, but replaces the covariance-weighted recursive update with a fixed-gain correction. Its correction is

$$\delta\hat{\boldsymbol{\lambda}}_k = \gamma_0 \boldsymbol{D}(\boldsymbol{H}_k\boldsymbol{D})^\dagger \tilde{\boldsymbol{e}}_k, \tag{42}$$

where $\boldsymbol{D} = \mathrm{diag}(\boldsymbol{I}_3, \ell\boldsymbol{I}_3)$, $\ell = \|{}^c\boldsymbol{\rho}_k^d\|_2 / \|{}^c\boldsymbol{\phi}_k^d\|_2$, and $(\cdot)^\dagger$ denotes the truncated pseudoinverse. The fixed gain $\gamma_0 = 0.03$ was determined using an exponential residual-decay criterion

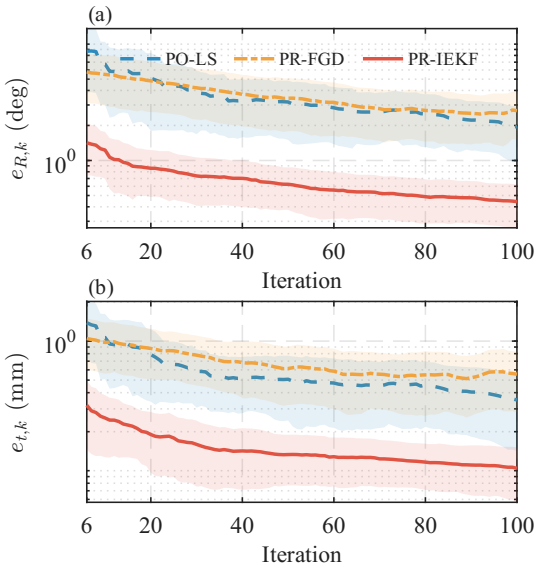


Fig. 2. Convergence comparison of the three calibration methods. Lines indicate the mean, and shaded regions represent the mean ± one standard deviation. The vertical axes are shown on logarithmic scales. (a) Rotation error. (b) Translation error.

and kept constant throughout all simulations. The correction is applied on $SE(3)$.

Let ${}^c\boldsymbol{R}_b^*$ and ${}^c\boldsymbol{t}_b^*$ denote the ground-truth hand-eye parameters. The rotation and translation errors are defined as

$$e_{R,k} = \left\| \left(\log\left({}^c\boldsymbol{R}_b^* ({}^c\hat{\boldsymbol{R}}_{b,k})^\top \right) \right)^\vee \right\|_2, \tag{43}$$

and

$$e_{t,k} = \left\| {}^c\boldsymbol{t}_b^* - {}^c\hat{\boldsymbol{t}}_{b,k} \right\|_2. \tag{44}$$

The final rotation and translation errors are denoted by e_R and e_t, respectively.

B. Convergence Comparison

Fig. 2 compares the convergence of the three methods. PR-IEKF achieves the lowest mean rotation and translation errors throughout the recursive updates, reaching $0.44 \pm 0.18°$ and 0.104 ± 0.047 mm after 100 motion increments. Compared with PO-LS, the mean rotation and translation errors are reduced by 77.2% and 70.3%, respectively.

The convergence of PR-FGD, which uses the same position residual and analytical Jacobian as PR-IEKF, demonstrates the effectiveness of the derived Jacobian. The lower estimation errors achieved by PR-IEKF relative to PR-FGD demonstrate the benefit of incorporating measurement covariance into the recursive update. Compared with PO-LS, which requires a subsequent projection of the rotation estimate onto $SO(3)$, PR-IEKF achieves lower estimation errors by accounting for anisotropic measurement noise and updating the hand-eye transformation directly on the $SE(3)$ manifold.

C. Sensitivity to Initialization and Excitation

To evaluate the combined influence of initialization and motion excitation, the initial rotation and translation error

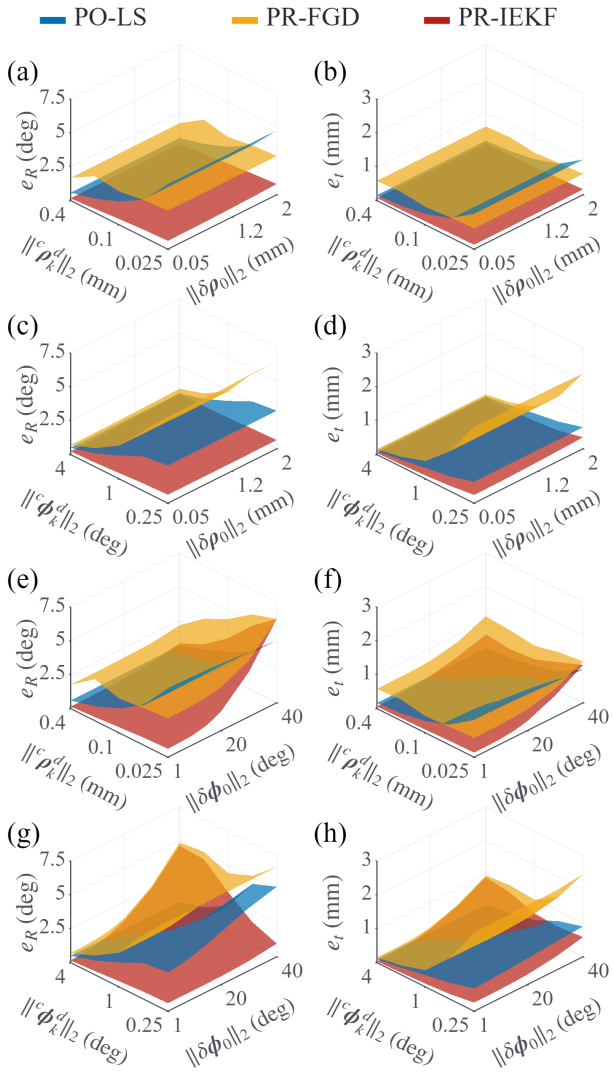


Fig. 3. Sensitivity to initialization and excitation. Columns show the mean final rotation and translation errors. Rows correspond to $(\|\delta\boldsymbol{\rho}_0\|_2, \|{}^c\boldsymbol{\rho}_k^d\|_2)$, $(\|\delta\boldsymbol{\rho}_0\|_2, \|{}^c\boldsymbol{\phi}_k^d\|_2)$, $(\|\delta\boldsymbol{\phi}_0\|_2, \|{}^c\boldsymbol{\rho}_k^d\|_2)$, and $(\|\delta\boldsymbol{\phi}_0\|_2, \|{}^c\boldsymbol{\phi}_k^d\|_2)$, respectively. The initial-error and final-error axes are linear, whereas the excitation-magnitude axes are logarithmic. All four panels in each column use the same final-error axis range.

magnitudes, $\|\delta\boldsymbol{\phi}_0\|_2$ and $\|\delta\boldsymbol{\rho}_0\|_2$, were varied together with the rotational and translational excitation magnitudes, $\|{}^c\boldsymbol{\phi}_k^d\|_2$ and $\|{}^c\boldsymbol{\rho}_k^d\|_2$. The tested ranges were $\|\delta\boldsymbol{\phi}_0\|_2 = 1°$ to $40°$, $\|\delta\boldsymbol{\rho}_0\|_2 = 0.05$ to 2.00 mm, $\|{}^c\boldsymbol{\phi}_k^d\|_2 = 0.25°$ to $4°$, and $\|{}^c\boldsymbol{\rho}_k^d\|_2 = 0.025$ to 0.40 mm. The remaining parameters were kept fixed.

Figs. 3(a)–(d) show the effects of initial translation error on calibration accuracy under different motion excitation settings. PR-IEKF maintains the lowest mean rotation and translation errors across all tested conditions, with relatively small variations as the initial translation error increases. This robustness is attributed to the recursive correction of the hand-eye transformation using the position residual and analytical Jacobian, while incorporating anisotropic measurement covariance into the update.

For variations in the initial rotation error, PR-IEKF achieves lower mean rotation and translation errors under most tested

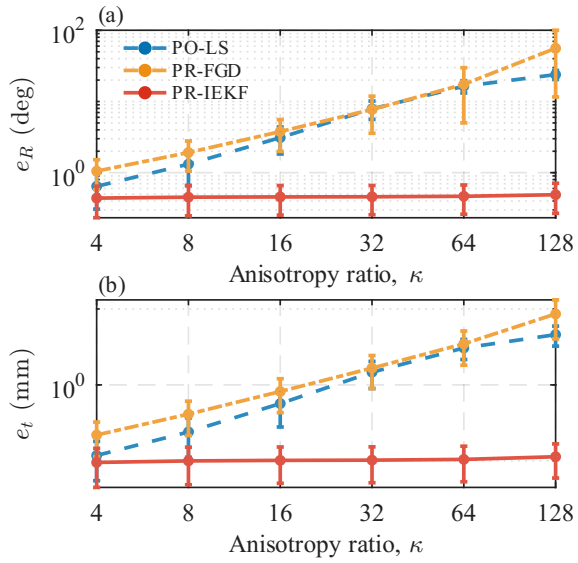


Fig. 4. Sensitivity of the three calibration methods to measurement anisotropy with fixed lateral noise and varying axial noise ($4 \leq \kappa \leq 128$). Markers indicate the mean final error, and error bars represent $\pm$ one standard deviation. Both axes are shown on logarithmic scales. (a) Rotation error. (b) Translation error.

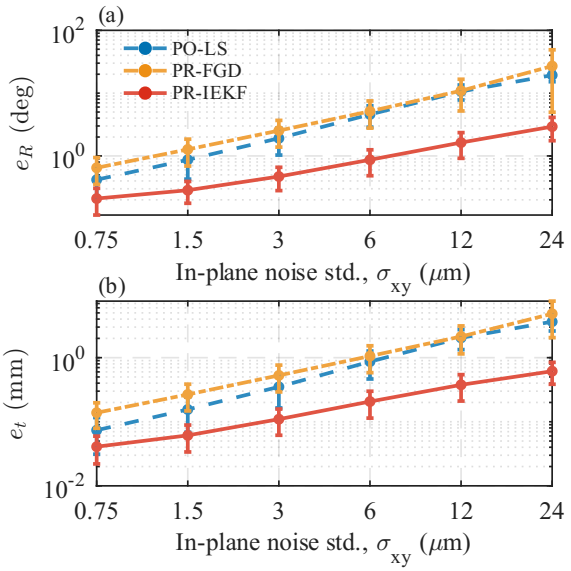


Fig. 5. Sensitivity of the three calibration methods to the measurement noise level at $\kappa = 10$. Markers indicate the mean final error, and error bars represent $\pm$ one standard deviation. Both axes use logarithmic scales. (a) Rotation error. (b) Translation error.

excitation settings, particularly at small and moderate initial rotation errors. As shown in Figs. 3(e)–(h), PR-IEKF maintains its accuracy advantage over PO-LS and PR-FGD across a wide range of initial rotation errors. This performance is supported by the analytical Jacobian, which relates the position residual to the hand-eye transformation error, and the recursive update that incorporates anisotropic measurement covariance.

D. Robustness to Anisotropic Noise

This simulation evaluates the effect of measurement anisotropy on calibration accuracy. The lateral noise standard deviations were fixed at $\sigma_x = \sigma_y = 0.003$ mm, while σ_z was varied. The in-plane noise standard deviation is defined as $\sigma_{xy} = \sqrt{(\sigma_x^2 + \sigma_y^2)/2}$, and the anisotropy ratio is defined as

$$\kappa = \frac{\sigma_z}{\sigma_{xy}}. \tag{45}$$

The tested values were $\kappa \in \{4, 8, 16, 32, 64, 128\}$, covering the anisotropy observed in the experimental measurements reported in Section V.

As shown in Fig. 4, PR-IEKF maintains the lowest mean rotation and translation errors across all tested anisotropy ratios. As κ increases from 4 to 128, its mean rotation and translation errors increase by only 11.5% and 18.8%, respectively, whereas the corresponding errors of PO-LS increase by approximately 37 and 39 times. This robustness is attributed to the incorporation of anisotropic measurement covariance into the recursive update, which reduces the influence of the less reliable axial measurements on the hand-eye estimation.

E. Sensitivity to Noise Level

To evaluate the influence of measurement noise level, the anisotropy ratio was fixed at $\kappa = 10$, while the in-plane noise standard deviation was varied as $\sigma_{xy} \in \{0.75, 1.5, 3, 6, 12, 24\}$ μm.

As shown in Fig. 5, the mean rotation and translation errors increase with measurement noise for all three methods, but PR-IEKF maintains the lowest errors across all tested noise levels. Compared with PO-LS, PR-IEKF reduces the mean rotation and translation errors by 49.9% and 44.7% at $\sigma_{xy} = 0.75$ μm, with the reductions increasing to 84.8% and 83.1% at $\sigma_{xy} = 24$ μm, respectively. The increasing advantage of PR-IEKF is attributed to the incorporation of measurement covariance into the recursive update, which accounts for measurement uncertainty as the noise level increases.

F. Summary

The simulations demonstrate that PR-IEKF achieves lower rotation and translation errors than PO-LS and PR-FGD, with smaller variations across repeated runs. PR-IEKF also maintains its accuracy advantage under varying initialization and motion excitation conditions. Furthermore, its estimation errors increase more slowly as measurement anisotropy and noise level increase, demonstrating improved robustness to measurement noise.

V. Experiments

This section evaluates PR-IEKF on a micromanipulation platform. The position measurement noise is first characterized for each optical configuration. Repeatability is evaluated under the $5\times$ configuration, and calibration performance is further examined using $5\times$, $10\times$, and $20\times$ objectives.

A. Experimental Setup

Fig. 6 shows the experimental platform, consisting of a PI H-811 six-axis parallel stage, a monocular microscope with a

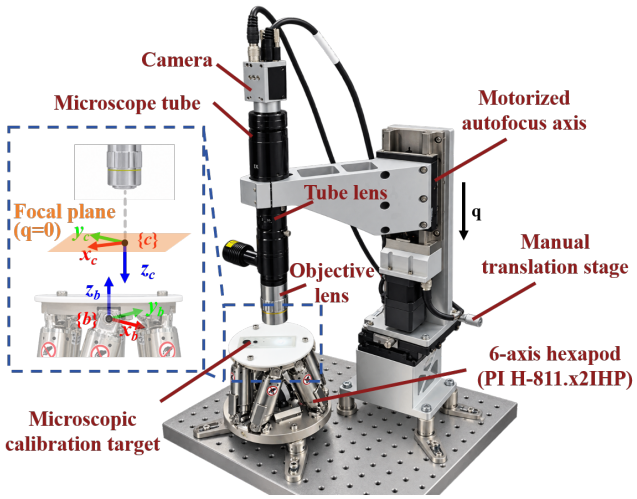


Fig. 6. Experimental platform for microscopic hand-eye calibration.

Basler acA2440-20gc camera, and a motorized autofocus axis. Without loss of generality, a circular marker on the calibration target was selected for tracking under the microscope. The lateral position (x_c, y_c) was obtained through image localization, and the axial position z_c through autofocus, as described in Section II-B. A fixed $2\times$ tube lens was used throughout the experiments.

For each optical configuration, the image scale was calibrated using known stage translations, and $\boldsymbol{\Sigma}_p$ was estimated from 50 repeated measurements of a stationary marker. The measurement statistics are summarized in Table I. Eight calibrations were performed per configuration, each with $N = 20$ motion increments. The stacked Jacobian from the first six increments had rank six, satisfying (31). Each validation sequence contained $N_v = 10$ increments.

The initial hand-eye estimate was obtained by applying rotational and translational offsets of 5° and 2 mm to a coarse calibration result, and was kept identical across methods and repeated runs within each configuration. For PR-IEKF, the per-axis prior standard deviations were 5° in rotation and 1 mm in translation. For PR-FGD, the fixed gain $\gamma_0 = 0.03$ was retained from the simulations.

For validation, the measured position residual is defined as

$$\boldsymbol{r}_k = {}^c\tilde{\boldsymbol{p}}_{k+1} - {}^c\hat{\boldsymbol{p}}_{k+1}, \tag{46}$$

where $\boldsymbol{r}_k = [r_{x,k}, r_{y,k}, r_{z,k}]^\top$. The lateral motion-consistency error is defined as

$$e_{xy} = \sqrt{\frac{1}{N_v}\sum_{k=0}^{N_v-1}\left(r_{x,k}^2 + r_{y,k}^2\right)}, \tag{47}$$

and the covariance-weighted motion-consistency error is

$$e_{\Sigma} = \sqrt{\frac{1}{3N_v}\sum_{k=0}^{N_v-1}\boldsymbol{r}_k^\top \boldsymbol{\Sigma}_{r,k}^{-1}\boldsymbol{r}_k}, \tag{48}$$

where $\boldsymbol{\Sigma}_{r,k}$ is computed according to (37) using the measured $\boldsymbol{\Sigma}_p$ and the validation motion. The metrics e_{xy} and e_{Σ} quantify the lateral and covariance-weighted 3-D motion consistency, respectively, with e_{Σ} being dimensionless.

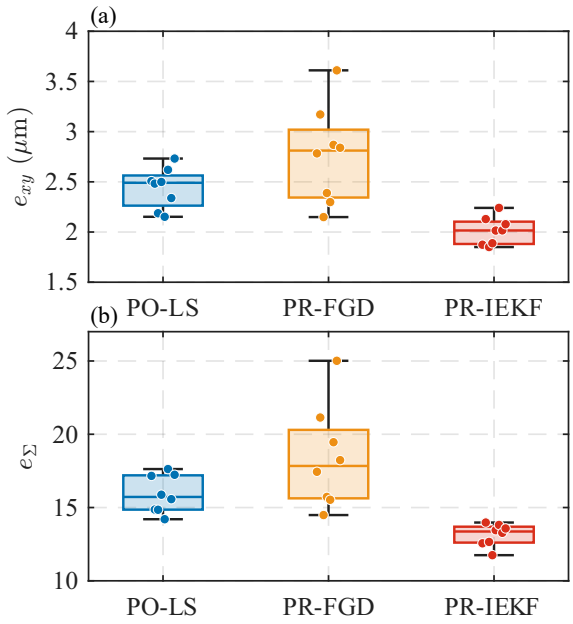


Fig. 7. Validation performance over eight repeated calibrations with the $5\times$ objective. Dots denote individual runs. (a) Lateral motion-consistency error e_{xy}. (b) Covariance-weighted motion-consistency error e_{Σ}.

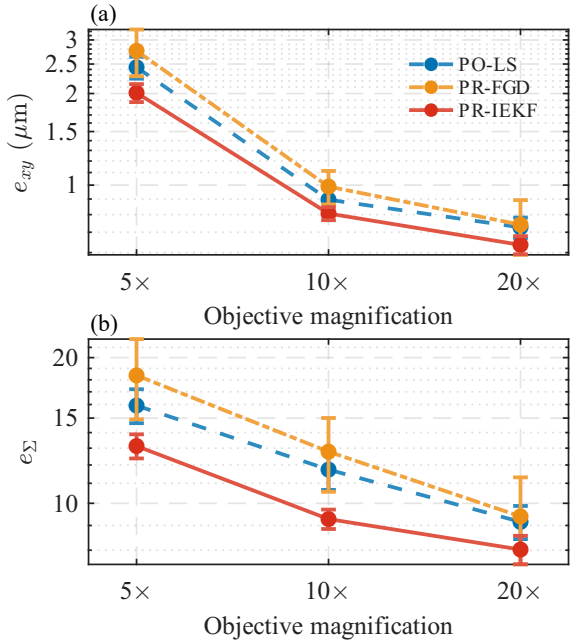


Fig. 8. Validation performance across three objective magnifications. Markers indicate the mean, and error bars represent ± one standard deviation over eight runs. (a) Lateral motion-consistency error e_{xy}. (b) Covariance-weighted motion-consistency error e_{Σ}.

B. Evaluation of Repeated Calibrations

Eight repeated calibration experiments were conducted using the $5\times$ objective to evaluate the consistency of calibration performance. The calibration settings and measurement

TABLE I
EXPERIMENTAL DETAILS FOR THREE OBJECTIVE MAGNIFICATIONS. BEST RESULTS ARE SHOWN IN BOLD.

Objective	Scale (nm/pixel)	Calibration step		Static observation statistics				Methods	Mean ± SD		Error Increase*	
		Translation (μm)	Rotation (deg)	σ_x (nm)	σ_y (nm)	σ_z (nm)	κ		e_{xy} (μm)	e_Σ	e_{xy}	e_Σ
5×	686.52	300	2	48.8	46.4	3200.7	67.22	PO-LS	2.44 ± 0.20	15.92 ± 1.28	↑ 21.4%	↑ 21.2%
								PR-FGD	2.76 ± 0.48	18.38 ± 3.46	↑ 37.3%	↑ 40.0%
								PR-IEKF	$\mathbf{2.01 \pm 0.14}$	$\mathbf{13.13 \pm 0.75}$	—	—
10×	338.64	150	1	26.5	38.4	620.8	18.83	PO-LS	0.90 ± 0.08	11.74 ± 1.08	↑ 11.1%	↑ 26.5%
								PR-FGD	0.99 ± 0.12	12.79 ± 2.22	↑ 22.2%	↑ 37.8%
								PR-IEKF	$\mathbf{0.81 \pm 0.04}$	$\mathbf{9.28 \pm 0.43}$	—	—
20×	166.56	75	0.5	32.0	33.0	312.5	9.62	PO-LS	0.73 ± 0.06	9.15 ± 0.72	↑ 14.1%	↑ 14.1%
								PR-FGD	0.74 ± 0.15	9.40 ± 1.92	↑ 15.6%	↑ 17.2%
								PR-IEKF	$\mathbf{0.64 \pm 0.04}$	$\mathbf{8.02 \pm 0.54}$	—	—

*Error increase relative to PR-IEKF, computed as $(e_{\text{baseline}} - e_{\text{PR-IEKF}})/e_{\text{PR-IEKF}} \times 100\%$.

statistics are provided in Table I. The validation sequence used translational and rotational step sizes of 450 μm and 3°, respectively.

As shown in Fig. 7, PR-IEKF achieves the lowest mean e_{xy} and e_Σ, with reductions of 17.6% and 17.5% relative to PO-LS, respectively. It also exhibits smaller standard deviations than both PO-LS and PR-FGD, indicating more consistent calibration performance across repeated runs. This performance is consistent with the incorporation of anisotropic measurement covariance into the recursive update, which reduces the influence of the less reliable axial measurements. Detailed error statistics are reported in Table I.

C. Validation under Different Objective Magnifications

The method was further evaluated with 10× and 20× objectives, in addition to the 5× configuration evaluated above. The experimental settings and detailed calibration results for all three objectives are summarized in Table I. The validation sequence used translational step sizes of 225 and 112.5 μm and rotational step sizes of 1.5° and 0.75° for the 10× and 20× objectives, respectively.

As shown in Fig. 8, PR-IEKF achieves the lowest mean e_{xy} and e_Σ across all three objectives. Despite variations in image scale and measurement anisotropy, PR-IEKF maintains its accuracy advantage by incorporating the corresponding measurement covariance into the recursive update.

D. Summary

The physical experiments demonstrate that PR-IEKF achieves lower mean motion-consistency errors and smaller standard deviations than PO-LS and PR-FGD. These advantages are maintained across three objective magnifications, with measured anisotropy ratios ranging from 9.62 to 67.22. The results further confirm the accuracy and robustness advantages observed in the simulations.

VI. CONCLUSION

This article proposes PR-IEKF for 6-DOF hand-eye calibration in micromanipulation using successive 3-D position measurements of a single tracked feature without visual orientation measurements. An analytical Jacobian relates the position residual to the hand-eye transformation error, enabling full 6-DOF estimation under sufficient motion excitation. PR-IEKF incorporates anisotropic position measurement covariance into recursive estimation and updates the hand-eye transformation directly on the $SE(3)$ manifold, without requiring subsequent rotation projection.

Simulations demonstrate that PR-IEKF achieves lower rotation and translation errors than PO-LS and PR-FGD, with greater robustness to initialization errors and measurement noise. Real-world experiments further show lower motion-consistency errors and smaller variations across repeated calibrations, while maintaining its accuracy advantage across different objective magnifications.

Future work will investigate adaptive measurement-noise estimation and validation under more diverse operating conditions.

REFERENCES

[1] X. Sha, H. Sun, Y. Zhao, W. Li, and W. J. Li, "A review on microscopic visual servoing for micromanipulation systems: Applications in micromanufacturing, biological injection, and nanosensor assembly," *Micromachines*, vol. 10, no. 12, Dec. 2019, art. no. 843.

[2] I. Enebuse, M. Foo, B. S. K. K. Ibrahim, H. Ahmed, F. Supmak, and O. S. Eyobu, "A comparative review of hand-eye calibration techniques for vision guided robots," *IEEE Access*, vol. 9, pp. 113 143–113 155, 2021.

[3] L. Ranno, P. Gupta, K. Gradkowski *et al.*, "Integrated photonics packaging: Challenges and opportunities," *ACS Photonics*, vol. 9, no. 11, pp. 3467–3485, Nov. 2022.

[4] B. Tamadazte, E. Marchand, S. Dembélé, and N. Le Fort-Piat, "CAD model-based tracking and 3D visual-based control for MEMS microassembly," *Int. J. Robot. Res.*, vol. 29, no. 11, pp. 1416–1434, Sep. 2010.

[5] C. R. Copeland, C. D. McGray, B. R. Ilic, J. Geist, and S. M. Stavis, "Accurate localization microscopy by intrinsic aberration calibration," *Nat. Commun.*, vol. 12, no. 1, Jun. 2021, art. no. 3925.

[6] H. Wang, K. Bai, J. Cui, Q. Shi, T. Sun, Q. Huang, P. Dario, and T. Fukuda, "Three-dimensional autofocusing visual feedback for automated rare cells sorting in fluorescence microscopy," *Micromachines*, vol. 10, no. 9, Aug. 2019, art. no. 567.

[7] R. Tsai and R. Lenz, "A new technique for fully autonomous and efficient 3D robotics hand/eye calibration," *IEEE Trans. Robot. Autom.*, vol. 5, no. 3, pp. 345–358, Jun. 1989.

[8] F. C. Park and B. J. Martin, "Robot sensor calibration: Solving AX=XB on the Euclidean group," *IEEE Trans. Robot. Autom.*, vol. 10, no. 5, pp. 717–721, Oct. 1994.

[9] K. Daniilidis, "Hand-eye calibration using dual quaternions," *Int. J. Robot. Res.*, vol. 18, no. 3, pp. 286–298, Mar. 1999.

[10] J. Ha, D. Kang, and F. C. Park, "A stochastic global optimization algorithm for the two-frame sensor calibration problem," *IEEE Trans. Ind. Electron.*, vol. 63, no. 4, pp. 2434–2446, Apr. 2016.

[11] Y. Motai and A. Kosaka, "Hand-eye calibration applied to viewpoint selection for robotic vision," *IEEE Trans. Ind. Electron.*, vol. 55, no. 10, pp. 3731–3741, Oct. 2008.

[12] X. Liu, H. Madhusudanan, W. Chen, D. Li, J. Ge, C. Ru, and Y. Sun, "Fast eye-in-hand 3-D scanner-robot calibration for low stitching errors," *IEEE Trans. Ind. Electron.*, vol. 68, no. 9, pp. 8422–8432, Sep. 2021.

[13] Z. Zhou, L. Ma, X. Liu, Z. Cao, and J. Yu, "Simultaneously calibration of multi hand-eye robot system based on graph," *IEEE Trans. Ind. Electron.*, vol. 71, no. 5, pp. 5010–5020, May. 2024.

[14] S. Qiu, M. Wang, and M. R. Kermani, "A new formulation for hand–eye calibrations as point-set matching," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 9, pp. 6490–6498, Sep. 2020.

[15] H. Xie, W. Li, and H. Liu, "General geometry calibration using arbitrary free-form surface in a vision-based robot system," *IEEE Trans. Ind. Electron.*, vol. 69, no. 6, pp. 5994–6003, Jun. 2022.

[16] Z. Liu, X. Liu, Z. Cao, X. Gong, M. Tan, and J. Yu, "High precision calibration for three-dimensional vision-guided robot system," *IEEE Trans. Ind. Electron.*, vol. 70, no. 1, pp. 624–634, Jan. 2023.

[17] L. Wu and H. Ren, "Finding the kinematic base frame of a robot by hand-eye calibration using 3D position data," *IEEE Trans. Autom. Sci. Eng.*, vol. 14, no. 1, pp. 314–324, Jan. 2017.

[18] B. Grossmann and V. Krüger, "Continuous hand-eye calibration using 3D points," in *Proc. IEEE 15th Int. Conf. Ind. Informat. (INDIN)*, pp. 311–318, 2017.

[19] B. Tamadazte, N. Le Fort-Piat, and S. Dembélé, "Robotic micromanipulation and microassembly using monoview and multiscale visual servoing," *IEEE/ASME Trans. Mechatronics*, vol. 16, no. 2, pp. 277–287, Apr. 2011.

[20] J. W. Kim, S. Wei, P. Zhang, P. Gehlbach, J. U. Kang, I. Iordachita, and M. Kobilarov, "Towards autonomous retinal microsurgery using RGB-D images," *IEEE Robot. Autom. Lett.*, vol. 9, no. 4, pp. 3807–3814, Apr. 2024.

[21] S. Dembélé, J. Bert, B. Tamadazte, and N. Piat, "A trifocal transfer-based virtual microscope for robotic manipulation of MEMS components," *Int. J. Optomechatronics*, vol. 4, no. 4, pp. 342–361, 2010.

[22] M. Zhou, M. Hamad, J. Weiss *et al.*, "Towards robotic eye surgery: Marker-free, online hand-eye calibration using optical coherence tomography images," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3944–3951, Oct. 2018.

[23] D. Zhang, F. P.-W. Lo, J.-Q. Zheng, W. Bai, G.-Z. Yang, and B. Lo, "Data-driven microscopic pose and depth estimation for optical microrobot manipulation," *ACS Photonics*, vol. 7, no. 11, pp. 3003–3014, Nov. 2020.

[24] A. Barrau and S. Bonnabel, "The invariant extended Kalman filter as a stable observer," *IEEE Trans. Autom. Control*, vol. 62, no. 4, pp. 1797–1812, Apr. 2017.

[25] Y. Chen, Y. Wang, F. Bu, G. Xin, C. Dai, X. Liu, Y. Sun, and X. Liu, "A fifth-order POE-based method for kinematic identification and inverse kinematics of serial robots," *IEEE Trans. Robot.*, vol. 42, pp. 1068–1087, 2026.

[26] S. Pertuz, D. Puig, and M. A. Garcia, "Analysis of focus measure operators for shape-from-focus," *Pattern Recognit.*, vol. 46, no. 5, pp. 1415–1432, May. 2013.

[27] J. Knapper, J. T. Collins, J. Stirling, S. McDermott, W. Wadsworth, and R. W. Bowman, "Fast, high-precision autofocus on a motorised microscope: Automating blood sample imaging on the OpenFlexure microscope," *J. Microsc.*, vol. 285, no. 1, pp. 29–39, Jan. 2022.

[28] T. D. Barfoot, *State Estimation for Robotics*. Cambridge, U.K.: Cambridge University Press, 2017.

[29] F. Chaumette and S. Hutchinson, "Visual servo control, part i: Basic approaches," *IEEE Robot. Autom. Mag.*, vol. 13, no. 4, pp. 82–90, 2006.



Haoran Cui received the B.Eng. degree in intelligent manufacturing engineering from Shandong University of Science and Technology, Qingdao, China, in 2024. He is currently working toward the Ph.D. degree with the Institute of Robotics and Intelligent Systems, Dalian University of Technology, Dalian, China. His research interests include visual servoing, robotics, and microassembly.



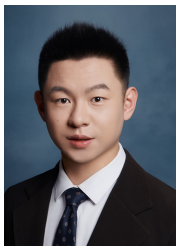
Weichen Dou received the B.Eng. degree in mechanical design, manufacturing, and automation from the School of Mechanical Engineering, Dalian University of Technology, Dalian, China, in 2025. He is currently pursuing the M.Eng. degree at the Institute of Robotics and Intelligent Systems, Dalian University of Technology. His current research interests include robotics, visual servoing, and the integration of artificial intelligence into autonomous systems.



Jia Shi received the B.Eng. degree in mechanical engineering from Dalian University of Technology, Dalian, China, in 2024. She is currently working toward the M.S. degree in mechanical engineering with the Institute of Robotics and Intelligent Systems, Dalian University of Technology, Dalian, China. Her research interests include piezoelectric stick-slip actuators and dynamic modeling.



Bowen Wang received the B.Eng. degree in mechanical design, manufacturing, and automation from the School of Mechanical Engineering, Dalian University of Technology, Dalian, China, in 2024. He is currently pursuing the M.Eng. degree at the Institute of Robotics and Intelligent Systems, Dalian University of Technology. His research interests lie in the fields of parallel manipulators, robot calibration, and microassembly.



Yuhan Chen (Member, IEEE) received the Ph.D. degree in mechanical engineering from the Beijing Institute of Technology, Beijing, China, in 2022 and then worked as a postdoctoral fellow at the Southern University of Science and Technology. He has been an assistant professor at the Institute of Robotics and Intelligent Systems at Dalian University of Technology since 2024. His research interests include visual servoing, robotics, and microassembly.



Xinyu Liu (Member, IEEE) received the B.Eng. and M.Eng. degrees from the Harbin Institute of Technology, Harbin, China, in 2002 and 2004, respectively, and the Ph.D. degree from the University of Toronto, Toronto, ON, Canada, in 2009, all in mechanical engineering. He is a Professor with the Department of Mechanical and Industrial Engineering, University of Toronto, and also with the Institute of Robotics and Intelligent Systems, Dalian University of Technology. Before joining the University of Toronto in September 2017, he was an Associate Professor and the Canada Research Chair in Microfluidics and BioMEMS with the Department of Mechanical Engineering, McGill University, Montreal, QC, Canada. His research interests include microrobotics, nanorobotics, soft robotics, and microfluidics.